

AGASTYA KATARIA

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PROFILE SUMMARY

Computer Science undergraduate at **Trinity College Dublin** with **International Mathematical Olympiad** achievements and hands-on experience building software across backend engineering, machine learning, cybersecurity, and quantitative finance projects.

EDUCATION

B.A. in Computer Science, Trinity College Dublin — First Class Honours (Distinction) (81%) | Sep 2024 – May 2028 (EGD)

Diploma in Python Programming, EBVTR — A Grade | Dec 2022 – Feb 2023

SAT — 1440 (780 Math, 660 Reading & Writing) | Feb 2023

KEY SKILLS EXPERIENCE

Languages: Python, Java, SQL

Frameworks & Libraries: Spring Boot, FastAPI, Flask, React, Pandas, NumPy, SciPy, Matplotlib, Plotly

Databases & Infrastructure: PostgreSQL, Docker, Kubernetes, Kafka, Prometheus

Security & Observability: Elasticsearch, Logstash, Kibana

Tools: Git, Linux, Gradle

Software Engineering: REST APIs, Data Structures & Algorithms, Distributed Systems, Machine Learning (LSTM, PPO)

Finance: FIX Protocol, Value-at-Risk (Historical & Monte Carlo), Financial Modelling

EXPERIENCE

Cybersecurity Engineering Intern | CRIS (Centre for Railway Information Systems) | May 2026 – Aug 2026

Built a Security Information and Event Management (SIEM) platform using Elasticsearch, Logstash and Kibana to collect and analyze logs from Windows and Linux systems.

Developed detection pipelines for brute-force attacks, privilege escalation and malware execution, with interactive Kibana dashboards and automated incident reporting.

Designed a machine-learning phishing detection system using URL features and email metadata, integrating real-time browser protection.

Developed PhishGuard, an open-source phishing simulation platform supporting campaign management, user interaction tracking and automated security awareness reporting.

PROJECTS

Rollout Feature Flag Tool- Toast (Backend Developer) | Jan 2026 – April 2026

Developing the backend of a team-built **feature-flagging platform** that enables **controlled rollout and rollback of software features** through **rule-based access**. Implemented **CRUD services for feature flags and users**, along with **DTO layers** and **persistence models**. Developed the **user feature**, designing models with attributes such as **name, email, team prefixes, and accessible flags**, and implemented the related backend logic before integrating the user system with **authentication**.

Designed **percentage rollout logic using user-ID hashing into 100 bins** for **deterministic exposure control**.

Contributed to backend functionality for **logging feature activity** and began implementing **conditional flags and additional rollout mechanisms**. Currently building **REST APIs for flag evaluation and configuration** with **database integration** and **service-layer validation** as part of an ongoing 10-week team project.

QuantLab-Suite

Developed a modular Java/Python trading simulator to explore order execution, portfolio analytics, and risk modelling. Implemented support for multiple order types, FIX message simulation, portfolio performance tracking, and risk analysis including Historical and Monte Carlo Value-at-Risk, volatility estimation, and stress testing. Explored a microservice-based architecture using Docker, PostgreSQL, Kafka, and Prometheus for data processing and monitoring.

AtmosLens HoloViz Air-Quality Decision App

Developed an interactive Python application for visualizing Open-Meteo air-quality forecasts and providing location-based activity recommendations. Built dashboards using Panel, HoloViews, GeoViews, hvPlot, Datashader, DuckDB, and Xarray, implementing features such as health-profile configuration, exposure scoring, location search, and forecast updates. Added over 250 automated tests covering recommendation logic, data access, SQL integration, and core application functionality.

Flight Data Analyser

Designed and implemented a large-scale flight data analysis system processing **10,000+ entries** of real-world flight logs for pattern identification, route optimization, and pricing insights. Developed a Skyscanner-inspired Java application using Processing that features interactive UI for searching, filtering, and comparing flight options. Key work included data parsing, visualization dashboards, custom search algorithms, and performance-tuned rendering of results.

ACHIEVEMENTS

- International Mathematical Olympiad — Multiple medals; highest International Rank 13 (2015), other ranks 39 (2016), 52 (2020). Ranked in top 0.05% globally (est.).
- International Science Olympiad — Rank 27.
- Eurofest 2019 — 2nd place, Odyssey of the Mind.

LEADERSHIP & ACTIVITIES

- Member, Trinity Student Management Fund (AUM ~€700k)
- Volunteer, Youth Red Cross (District Level Camps)
- STEM Educator, Spread Smile Foundation (Academic mentoring)

PROFESSIONAL STRENGTHS

- Analytical Problem Solving — Strong mathematical reasoning developed through International Mathematical Olympiad competitions and applied to software engineering projects.
- Adaptability — Quickly learn and apply new technologies across machine learning, cybersecurity, and distributed systems.
- Collaboration — Communicate technical ideas effectively and contribute to team-based software projects.

COURSES & CERTIFICATIONS

Mathematics for Machine Learning (Imperial, Coursera) | Data Structures & Algorithms (UCSD, Coursera) | Finance & Quantitative Modeling (Wharton Online) | Cybersecurity Fundamentals (IBM) | Introduction to CS & Programming using Python (MITx) | Financial Markets (Yale) | Web Dev & API Specialization (Coursera)
